Autocorrelation Analysis in Florida weather

Bridget Smith, Anna Cavalieri Canosa, Lehan Geng, Zhilu Zhang

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1 Introduction

This document presents the results of the autocorrelation analysis of Key West annual mean temperatures. The goal of this analysis was to determine whether the temperatures of one year are significantly correlated with those of the following year across the entire time series.

2 Results & Conclusion

Results

The original correlation coefficient between successive years' temperatures was found to be 0.326. In the 10,000 simulations, only 6 of the random correlation coefficients were greater than the original correlation. The approximate p-value is therefore:

$$\text{p-value} = \frac{6}{10000} = 0.0006$$

Interpretation

The original correlation coefficient of 0.326169 indicates a positive but weak linear relationship between consecutive years. The p-value of 0.0006 is much smaller than the commonly used significance level of 0.05, indicates that the original correlation is statistically significant and unlikely to have occurred by chance.

Conclusion

The results of this analysis demonstrates that temperatures of one year are significantly correlated with those of the next year in the given location.